Databases: Counting, Grouping and Joining Tables

Total things up and connect tables like a pro.

Databases | Beginner to Intermediate | Ages 12–18

InfiBee Coders

💡 The Big Idea

- Aggregate functions summarise many rows into one answer.
- `GROUP BY` makes one summary per group, like per class.
- A `JOIN` combines rows from two tables using a matching column.
- Joins let you keep data in neat separate tables instead of repeating it.

🧮 Aggregate Functions

- `COUNT(*)` counts rows.
- `SUM(col)` adds up numbers.
- `AVG(col)` gives the average.
- `MIN(col)` and `MAX(col)` give the smallest and largest.
- `ROUND(x, 1)` rounds to 1 decimal place.

👥 GROUP BY and HAVING

- `GROUP BY class` makes one result row per class.
- Use `HAVING` to filter groups. `WHERE` filters rows before grouping.
- Every selected column must be grouped or be an aggregate.

🔗 Joins

- A primary key is a unique id. A foreign key is a column that points to another table's key.
- `JOIN ... ON a.col = b.col` matches rows from two tables.
- `INNER JOIN` keeps only rows that match in both tables.
- `LEFT JOIN` keeps every row from the left table, even without a match.
- Use short aliases like `s` and `c` to keep queries tidy.

💻 Code Examples

Count, average and highest SQL

```sql
SELECT COUNT(*) AS students, ROUND(AVG(score), 1) AS average, MAX(score) AS top FROM students;
```

▶ **Output:**

```
students  average  top
--------  -------  ---
6         81.7     95
```

Average per class SQL

```sql
SELECT class, COUNT(*) AS students, ROUND(AVG(score), 1) AS average FROM students GROUP BY class;
```

▶ **Output:**

```
class  students  average
-----  --------  -------
6A     2         80.0
6B     1         90.0
7B     2         88.0
8C     1         64.0
```

One summary row per class.

Filter groups with HAVING SQL

```sql
SELECT class, ROUND(AVG(score), 1) AS average FROM students GROUP BY class HAVING AVG(score) > 85;
```

▶ **Output:**

```
class  average
-----  -------
6B     90.0
7B     88.0
```

Only classes with an average above 85.

See the classes table SQL

```sql
SELECT * FROM classes;
```

▶ **Output:**

```
class  teacher
-----  -------
6A     Ms Rao
6B     Mr Khan
7B     Ms Iyer
8C     Mr Das
```

Join two tables SQL

```sql
SELECT s.name, s.class, c.teacher FROM students s JOIN classes c ON s.class = c.class;
```

▶ Output:

```
name   class  teacher
-----  -----  -------
Aarav  6A     Ms Rao
Mia    7B     Ms Iyer
Zoya   6A     Ms Rao
Kabir  8C     Mr Das
Anaya  7B     Ms Iyer
Dev    6B     Mr Khan
```

Each student is matched with their class teacher.

Join with a filter and sort SQL

```sql
SELECT s.name, c.teacher FROM students s JOIN classes c ON s.class = c.class WHERE
s.score >= 85 ORDER BY s.name;
```

▶ Output:

```
name   teacher
-----  -------
Aarav  Ms Rao
Dev    Mr Khan
Mia    Ms Iyer
```

📚 Key Terms

Aggregate
A function that summarises many rows.

COUNT
Counts rows.

AVG
Average of numbers.

GROUP BY
Groups rows for summaries.

HAVING
Filters groups.

JOIN
Combines two tables.

ON
Says which columns must match.

Primary key
A unique id in a table.

Foreign key
A column that points to another table.

Alias
A short nickname for a table or column.

🧮 Quick Reference

```
SELECT COUNT(*) FROM t;
```

```
SELECT AVG(col) FROM t;
```

```
SELECT col, COUNT(*) FROM t GROUP BY col;
```

```
... HAVING COUNT(*) > 1
```

```
SELECT a.x, b.y FROM a JOIN b ON a.id = b.a_id;
```

✏️ Worked Examples

Q: How many students are in class 7B?
A: SELECT COUNT(*) FROM students WHERE class = '7B'; gives 2 (Mia and Anaya).

Q: What is the average score of class 6A?
A: SELECT AVG(score) FROM students WHERE class = '6A'; gives 80 because (88 + 72) / 2 = 80.

Q: Show each student with their teacher.
A: Join students to classes ON s.class = c.class and select s.name and c.teacher.

🚫 Common Mistakes

- Using `WHERE` to filter groups. Use `HAVING` instead.
- Forgetting `ON` in a join, which pairs every row with every row.
- Selecting a column that is not grouped when using `GROUP BY`.
- Not using aliases, making long queries hard to read.
- Joining on the wrong columns.

👨‍👩‍👧 Parent Corner

- Use a school example: count the kids in each class, then match each class with its teacher.
- Ask your child to explain a JOIN as matching names on two lists.
- Ask them to guess the result before running the query.
- Encourage plenty of practice with small tables.
- Celebrate when they can answer a real question with a query.

🎯 Quick Check

1. Which function counts rows?

✅ COUNT(*).

2. What does GROUP BY do?

✅ It groups rows so you get one summary per group.

3. What is the difference between WHERE and HAVING?

✅ WHERE filters rows before grouping. HAVING filters groups after.

4. What does a JOIN do?

✅ It combines rows from two tables using a matching column.

5. What does ON do in a join?

✅ It says which columns must match.